



Swathi Sri R 250701769-2025-CSE ▾

S2**Started on** Friday, 11 September 2026, 3:16 PM**State** Finished**Completed on** Friday, 11 September 2026, 3:20 PM**Time taken** 3 mins 52 secs**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 10.00 out of 10.00**Playing with Numbers:**

Ram and Sita are playing with numbers by giving puzzles to each other. Now it was Ram term, so he gave Sita a positive integer 'n' and two numbers 1 and 3. He asked her to find the possible ways by which the number n can be represented using 1 and 3. Write any efficient algorithm to find the possible ways.

Example 1:**Input:** 6**Output:** 6**Explanation:** There are 6 ways to 6 represent number with 1 and 3

1+1+1+1+1+1

3+3

1+1+1+3

1+1+3+1

1+3+1+1

3+1+1+1

Input Format

First Line contains the number n

Output Format**Print: The number of possible ways 'n' can be represented using 1 and 3**

Sample Input

6

Sample Output

6

Answer: (penalty regime: 0 %)

```

1 #include <stdio.h>
2
3 int main() {
4     int n;
5     if (scanf("%d", &n) != 1) return 0;
6
7     if (n < 3) {
8         printf("1\n");
9         return 0;
10    }
11
12    long long a = 1, b = 1, c = 1, current;
13    for (int i = 3; i <= n; i++) {
14        current = c + a;
15        a = b;
16        b = c;
17        c = current;
18    }
19
20    printf("%lld\n", c);
21    return 0;
22 }
23

```

	Input	Expected	Got	
✓	6	6	6	✓

	Input	Expected	Got	
✓	25	8641	8641	✓
✓	100	24382819596721629	24382819596721629	✓

Passed all tests! ✓

Correct

Marks for this submission: 10.00/10.00.

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